

RESEARCH THEMES

Science of deep learning

- Developed a mechanistic account of physical misgeneralization in sequence models for NeurIPS 2026.
- Worked on pre-training & in-context mechanisms w/ interpretable synthetic graph tasks (2023–2025).
- **First-author ICML 2025**; co-author **ICLR 2025** & **ICML 2024**; **first-author ICLR 2026** and co-author **NeurIPS 2023 & 2024** workshops.
- Supported by Harvard PRISE 2024 (\$3K + residency) & CBS-NTT Fellowship 2023 (\$5K).
- Advised by H. Tanaka & E. S. Lubana (PAI), H. Pfister (VCG), and K. Kumar & R. Tang (Comcast).

Multimodal modeling (audio & speech; computer vision)

- Analyzed spurious onsets in full-duplex speech systems at Comcast. Submitting to ICASSP 2027.
- Worked on multimodal speech and diffusion-based speech generation models (2025).
- Enhanced robust vision under partial/noisy supervision (2020–2024); **first-author CVPR 2024**.
- **First-author CVPR 2021 & AAAI 2021** at age 16.
- Advised by K. Kumar & R. Tang (Comcast), H. Pfister (VCG), and T. Höllerer (Four Eyes Lab).

AI governance, safety, and scientific applications

- **MATS 7.0** (winter 2025) research fellow; \$12K grant + \$10K compute.
- Mixed-methods study of ethics review across AI paper resubmissions for NeurIPS 2026 workshop.
- “Evolutionary Curriculum Learning Improves Biological Sequence Modeling,” **ICML 2026** workshop.

TECHNICAL PROJECTS

LiveTL Apps

- Founding dev of LiveTL, HyperChat, and YtcFilter, addons that improve YouTube & Twitch.
- 130K+ total users; 1,000+ repo stars; 30+ code contributors. Free, open-source, and cross-platform.

Torch Pitch Shift — PyPI

- The first Python library for pitch-shifting on GPU at the time; later added to PyTorch upstream.
- 1.7M+ downloads/month; 135+ GitHub stars; used by torch-audiomentations (1.2K+ stars).

Torch Time Stretch — PyPI

- GPU-accelerated PyTorch library for fast audio time-stretching and efficient transformation search.
- 40+ GitHub stars; used by ByteDance’s BuboGPT research code (500+ stars).

iframe Translator — npm

- A reverse-engineered hack of Google’s free in-browser translation feature, exploiting iframe loopholes.

slsh: SSH without keyboard lag

- Built a drop-in SSH wrapper that hides interactive latency using temporary local terminal predictions.
- Written in Rust; supports Linux, macOS, and Windows on x86-64 and ARM64.

holoEN Christmas Advent Calendar

- Full-stack dev of the event platform (holoen-advent.com); officially commissioned by Cover Corp.
- 250K+ viewers; a beloved yearly tradition in the hololive English community (2022, 2023, 2024).

SOUND & AUDIO TECHNOLOGY

Music composition/production

- Making original music since 2017; 20+ tracks w/ Reaper, VSTs, MIDI controllers, etc.
- Highlights: “HyperChat Trailer Theme” (2022, 16K views on YouTube), “Bliss” (2026, YouTube).

Plugins & tools

- Pogify: synchronized music playback extension. 10+ contributors; 50+ stars; 15K+ social media likes.
- ClipSynth: WebAssembly, Web Audio, WebUSB, Web MIDI tool to convert arbitrary clips into synths.

ALL PUBLICATIONS

Mechanisms of Misgeneralization in Physical Sequence Modeling

Kento Nishi, Raphael Tang, Karun Kumar, Core Francisco Park, Hidenori Tanaka

arXiv Preprint 2026, as **first author**.

- We show generated trajectories can look plausible while shifting aggregate distance or energy.
- We use a data deviation kernel to predict and reduce this form of physical misgeneralization.

Evolutionary Curriculum Learning Improves Biological Sequence Modeling

Richard Zhu, Kento Nishi

ICML 2026 SPIGM Workshop, as co-author.

- We gradually expand training from close evolutionary neighbors to distant homologs.
- This improves protein variant-effect prediction and RNA sequence generation.

When does Observational Data Teach Latent Dynamics?

Understanding Control Misalignment with Synthetic Tasks

Kento Nishi, Raphael Tang, Karun Kumar, Core Francisco Park, Hidenori Tanaka

ICLR 2026 Sci4DL Workshop, as **first author**.

- We show generated samples can fit data while violating latent controls like speed or energy.

Representation Shattering in Transformers: A Synthetic Study with Knowledge Editing

Kento Nishi, Rahul Ramesh, Maya Okawa, Mikail Khona, Hidenori Tanaka, Ekdeep Singh Lubana

ICML 2025, as **first author**.

- We show editing can distort untargeted entity representations and factual-recall geometries.

In-Context Learning of Representations

Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, Hidenori Tanaka

ICLR 2025, as co-author.

- We show concept representations can mediate learning from graph-structured examples.

Structured In-Context Task Representations

Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Kento Nishi, Maya Okawa, Hidenori Tanaka

NeurIPS 2024 NeurReps Workshop, as co-author.

- Precursor to ICLR 2025 “In-Context Learning of Representations.”

Towards an Understanding of Stepwise Inference in Transformers:

A Synthetic Graph Navigation Model

Mikail Khona, Maya Okawa, Jan Hula, Rahul Ramesh, Kento Nishi, Robert Dick, Ekdeep Singh Lubana, Hidenori Tanaka

ICML 2024, as co-author.

- We test stepwise inference, simplicity bias, and in-context primacy with graph navigation.

Stepwise Inference in Transformers: Exploring a Synthetic Graph Navigation Task

Mikail Khona, Maya Okawa, Rahul Ramesh, Kento Nishi, Robert P. Dick, Ekdeep Singh Lubana, Hidenori Tanaka

NeurIPS 2023 R0-FoMo Workshop, as co-author.

- Precursor to ICML 2024 “Towards an Understanding of Stepwise Inference in Transformers.”

Joint-Task Regularization for Partially Labeled Multi-Task Learning

Kento Nishi, Junsik Kim, Wanhua Li, Hanspeter Pfister

CVPR 2024, as **first author**.

- We regularize shared-space outputs to improve partially labeled multi-task learning at linear cost.

Augmentation Strategies for Learning with Noisy Labels

Kento Nishi, Yi Ding, Alex Rich, Tobias Höllerer

CVPR 2021, as **first author**.

- We separate loss-modeling and learning augmentations to improve noisy-label robustness.

Improving Label Noise Robustness with Data Augmentation and Semi-Supervised Learning

Kento Nishi, Yi Ding, Alex Rich, Tobias Höllerer

AAAI 2021 Student Abstract Track, as **first author**.

- Precursor to CVPR 2021 “Augmentation Strategies for Learning with Noisy Labels.”